THE BIG RIP

You can think of it as an unraveling.

The first things to go are the largest, most tenuously bound. Giant clusters of galaxies, in which groups of hundreds or thousands of galaxies flow lazily around each other in long intertwined paths, begin to find that those paths are growing longer. The wide spaces traversed by the galaxies over millions or billions of years widen even more, causing the galaxies at the fringes to slowly drift away into the growing cosmic voids. Soon, even the densest galaxy clusters find themselves inexorably dissipated, their component galaxies no longer feeling any central pull.

From a vantage point within our own galaxy, the loss of the clusters should be the first ominous sign that the Big Rip is in progress. But the speed of light delays this clue until we are already feeling the effects much closer to home. As our local cluster, Virgo, begins to dissipate, its previously languid motion away from the Milky Way begins to pick up speed. This effect is subtle, though. The next one is not.

We already have astronomical all-sky surveys that are capable of measuring the positions and motions of billions of stars within our own galaxy.[fn3] As the Big Rip approaches, we start to notice that the stars on the edges of the galaxy are not coming around in their expected orbits, but instead drifting away like guests at a party at the end of the evening. Soon after, our night sky begins to darken, as the great Milky Way swath across the sky fades. The galaxy is evaporating.

From this point, the destruction picks up its pace. We begin to find that the orbits of the planets are not what they should be, but are instead slowly spiraling outward. Just months before the end, after we've lost the outer planets to the great and growing blackness, the Earth drifts away from the Sun, and the Moon from the Earth. We too enter the darkness, alone.

The calm of this new solitude doesn't last.

At this point, any structure still intact is straining under the push of the expanding space within it. The Earth's atmosphere thins, from the top. Tectonic motions within the Earth respond chaotically to the shifting gravitational forces. With only hours to go, the Earth cannot hold together: our planet explodes.

Even the destruction of Earth, could, in principle, be survivable, if, having interpreted the signs, you have already retreated to some compact